Data Objects, Attribute Types & Statistical Descriptions

Dataset	HR_comma_sep.csv
Unit Coverage	Data Objects & Attribute Types and Basic Statistical Descriptions.
Tools	Power BI + Google Colab

Background :

The HR dataset contains 14,995 records of employee profiles, capturing metrics such as satisfaction level, last evaluation score, number of projects, average monthly hours, tenure, work accidents, whether the employee left the company, promotions in the last five years, department, and salary. This dataset is rich with quantitative, categorical, and binary attributes, making it ideal for statistical exploration and visualization.

Tasks :

PART A – Data Object Classification :

1. Quantitative Multidimensional Data

- satisfaction_level
- last_evaluation
- number_project
- average_montly_hours
- time_spend_company

Explanation:

These are continuous/numeric features representing employee metrics.

2. Categorical & Mixed Data :

- department
- salary Explanation:

These represent categories like job roles and salary levels.

3. Binary Data :

- left
- work_accident
- promotion_last_5years Explanation:

These have only two values (0/1), indicating yes/no conditions.

4. Dependency Analysis :

The HR dataset is non-dependency oriented because each employee record is independent. However, slight temporal dependency may exist due to tenure (time_spend_company), but no strong sequence or network relationship is present.

PART B – Central Tendency & Dispersion :

Code:

```
import pandas as pd import numpy as np

df = pd.read_csv("HR_comma_sep.csv")

cols =

['satisfaction_level','last_evaluation','number_project','average_montly_hours','time_spen

d_company'] summary = pd.DataFrame() summary['Mean'] = df[cols].mean()

summary['Median'] = df[cols].median() summary['Mode'] = df[cols].mode().iloc[0]

summary['Range'] = df[cols].max() - df[cols].min() summary['Variance'] = df[cols].var()

summary['Std Dev'] = df[cols].std() summary['IQR'] = df[cols].quantile(0.75) -

df[cols].quantile(0.25) summary
```

Output in Google colab :

	Mean	Median	Mode	Range	Variance	Std Dev	IQR
satisfaction_level	0.612890	0.64	0.10	0.91	0.061809	0.248615	0.38
last_evaluation	0.716108	0.72	0.55	0.64	0.029298	0.171166	0.31
number_project	3.803001	4.00	4.00	5.00	1.518478	1.232266	2.00
average_montly_hours	201.050150	200.00	135.00	214.00	2494.338821	49.943356	89.00
time_spend_company	3.498166	3.00	3.00	8.00	2.132366	1.460262	1.00

The statistical summary shows variation across all features. The average monthly hours and number of projects show higher dispersion, indicating variability in workload among employees. Satisfaction level is relatively less dispersed, suggesting consistency in employee satisfaction.

PART C – Univariate & Multivariate Analysis :

1. Univariate Analysis Code

Code : import seaborn as sns

import matplotlib.pyplot as plt

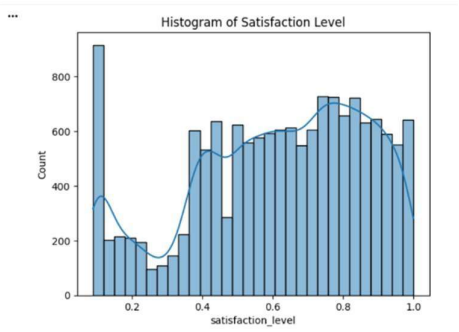
import scipy.stats as stats

Histogram sns.histplot(df['satisfaction_level'],

kde=True) plt.title("Histogram of Satisfaction

Level") plt.show()

Output in google colab:



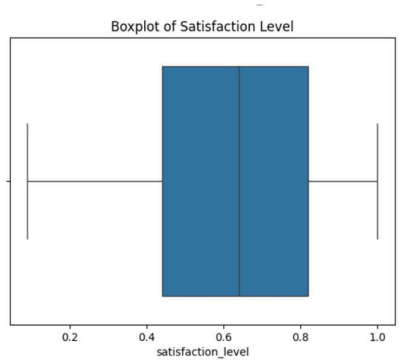
Boxplot :

Code :

sns.boxplot(x=df['satisfaction_level'])

plt.title("Boxplot of Satisfaction Level") plt.show()

Output in google colab :



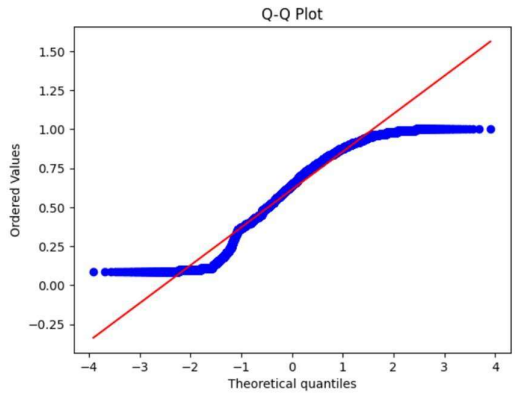
Q-Q plot Code

:

stats.probplot(df['satisfaction_level'], dist="norm", plot=plt)

plt.title("Q-Q Plot") plt.show()

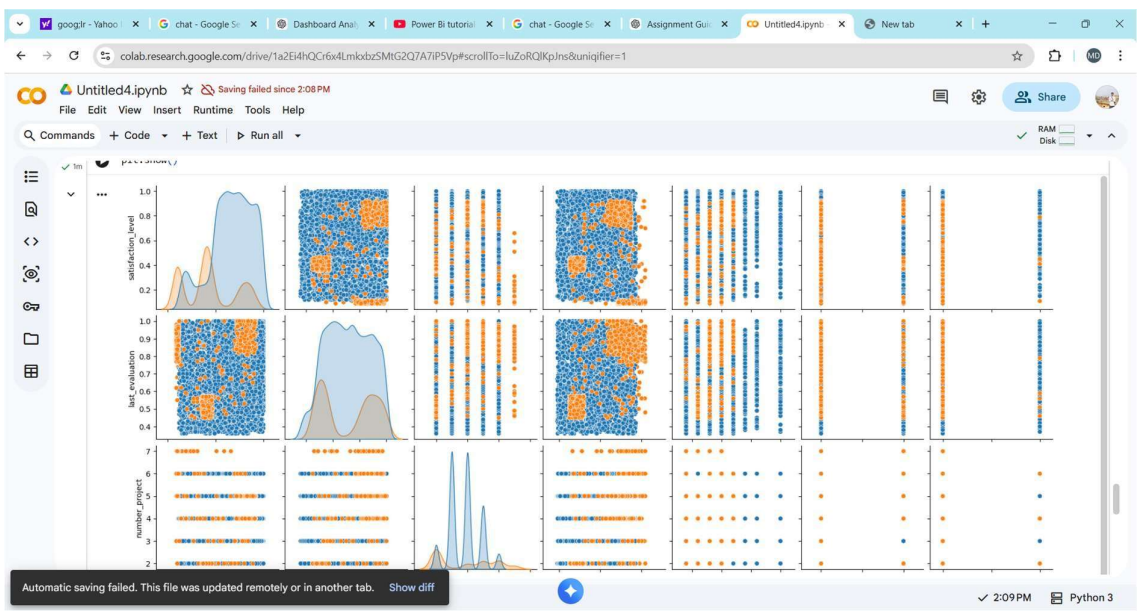
Output in google colab :



2. Pair Plot :

Code :

sns.pairplot(df, hue='left') plt.show()



The pair plot shows clear separation between employees who left and those who stayed. Features such as satisfaction level and average monthly hours show noticeable di erences between the two groups.

3. Correlation :

Code :

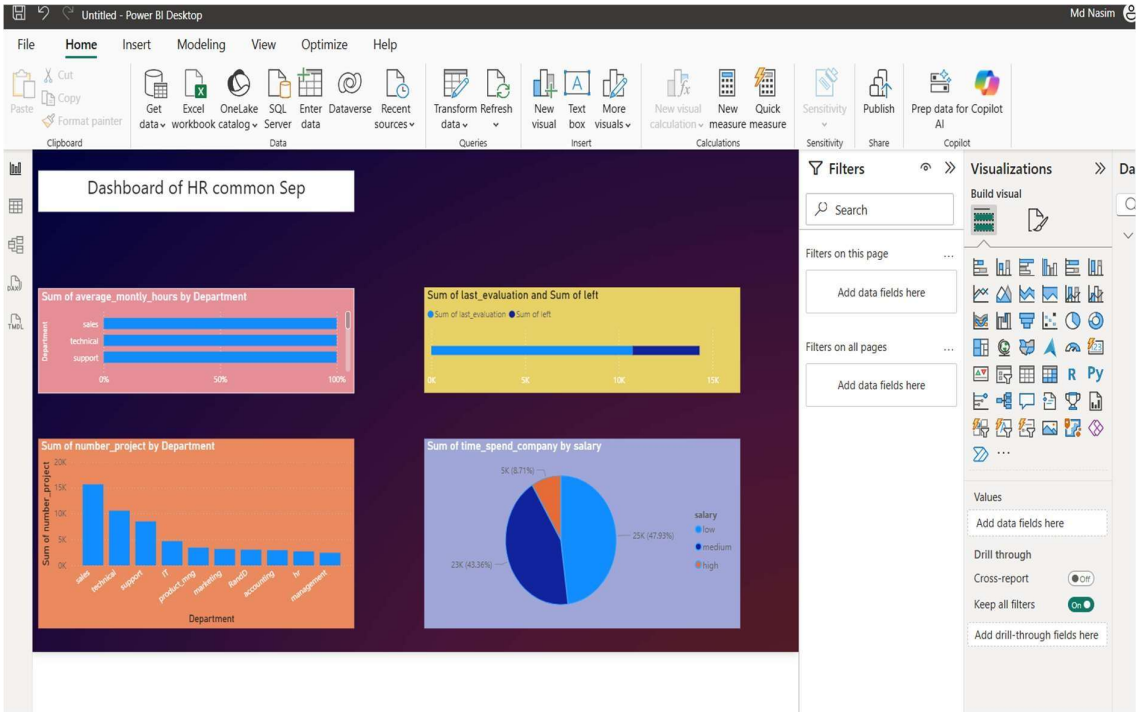
df['last_evaluation'].corr(df['number_project']) Output :

```
df['last_evaluation'].corr(df['number_project'])

np.float64(0.3490668805610896)
```

There is a moderate positive correlation between last evaluation and number of projects, indicating that employees working on more projects tend to have higher evaluation scores.

Dashboard of HR common Sep :



The dashboard reveals that employee attrition varies across departments, with certain departments showing higher turnover. Employees with low satisfaction levels are more likely to leave. Additionally, employees with high working hours and high evaluation scores also show higher chances of leaving, indicating possible work pressure.